

Structure over Scale: Hierarchy-Constrained Modeling for English Online Polarization Detection

Durga Nebhrajani
Team Ugh

Rakshit Aggarwal
Team Ugh

Manav Beriwal
Team Ugh

Abstract

SemEval-2026 Task 9 has a hierarchical label structure. Social dimensions are annotated only for texts already judged polarized, so a text is polarized exactly when some dimension applies. Shared-task systems ignore this. They train the subtasks separately, or share an encoder and little else, and the result is predictions that contradict the hierarchy (Guo and Chang, 2026). Maharjan et al. (2026) measured how often this happens, found rates as high as 33%, and removed the contradictions with a deterministic rule applied at inference. The rule cost them multi-label recall. Is that cost unavoidable, or does it come from fixing the problem too late? Our model computes the detection score from the social-dimension logits through a noisy-OR. Contradictory outputs then cannot be represented at all, and the detection loss reaches the dimension detectors even on text that is not polarized. Adopting the Logical Violation Rate of Maharjan et al. (2026), we compare independent, multi-task, gated, and structured models, and set macro-F1 against model size for the 12B–27B systems leading the English leaderboard.

1 Project Idea

This project takes the first two subtasks of SemEval-2026 Task 9, English only. POLARDETECT asks whether a text is polarized. POLARTYPE assigns one or more social dimensions (Political, Racial or Ethnic, Religious, Gender or Sexual Identity, and Other) to text that is polarized (Naseem et al., 2026).

The two subtasks are nested, not just related. POLARTYPE labels exist only where POLARDETECT is positive, so a text is polarized if and only if at least one dimension applies. Multi-task learning as usually practised puts independent heads on a shared encoder. Nothing there represents the constraint, and the model can end up predicting something logically impossible.

Our contributions are:

- (C1) A model whose POLARDETECT logit is computed from the POLARTYPE logits. Violations stop being something to clean up afterwards and become impossible to represent. Maharjan et al. (2026) do the opposite, detaching the tasks with a stop-gradient and reconciling them at inference; we couple them through the loss.
- (C2) A test of whether the recall cost Maharjan et al. (2026) report belongs to the constraint itself or to the point at which it is applied. Their Logical Violation Rate gives us the measure, extended here to the POLARDETECT/POLARTYPE pair and to both directions of violation.
- (C3) A comparison matched on compute. Our encoder has 184M parameters against the 12B to 27B of the systems holding the English state of the art, with macro-F1 reported against parameter count.

2 Motivation

Polarized language carries hostility, stereotyping, exclusion, or divisive framing, but these signals are contextual rather than lexical. Political vocabulary alone does not make a sentence polarized, polarized content is often oblique, and one text may involve several dimensions at once.

The task also has structure that current systems throw away. Guo and Chang (2026) train each subtask separately and note that “independent modeling causes hierarchical violations: some instances labeled non-polarized in Subtask 1 receive positive Subtask 2 or 3 labels.” They suggest post-hoc gating or hierarchical calibration as future work, implement neither, and give no numbers. Lin et al. (2026) fall back on an inference-time heuristic that

checks the POLARDETECT prediction before as-signing POLARTYPE labels.

Maharjan et al. (2026) take this further. They define the Logical Violation Rate, report values as high as 33%, and apply the rule $\hat{y}_{S1} = 0 \Rightarrow \hat{y}_{S2} = 0$ at inference. Their architecture feeds the child subtask detached parent logits, which by design stops the two tasks from shaping each other’s representations. The cost is stated openly in their paper: the constraint “slightly limits raw multi-label recall.” Their English numbers show it. Gating lifts POLARDETECT from 0.752 to 0.826 but drops POLARTYPE from 0.511 to 0.506.

All of these fixes operate on outputs, and all of them trade recall for consistency. That is what one would expect. Gating throws away dimension predictions whenever the detection head is wrong, so an error at the top of the hierarchy propagates downward. Computing detection from the dimension detectors avoids the problem, because nothing is thrown away. The constraint instead restricts what the model can learn, and on non-polarized text the detection loss reaches the dimension detectors directly, which is exactly where POLARTYPE supervision is missing.

3 Background

SemEval-2026 Task 9 introduces the POLAR benchmark, which covers 22 languages, and it attracted 67 teams and 73 system papers (Naseem et al., 2026). Only the English subset and the first two subtasks concern us here.

Two things about the English data shaped our design. Annotator agreement is only fair, at Fleiss’s $\kappa = 0.39$, well below Khmer (0.83) or Nepali (0.79), so some of the label signal is genuinely contested. Headroom also differs a great deal between the subtasks. The best English POLARDETECT score is 0.825 against a baseline of 0.773, while POLARTYPE reaches 0.532 against 0.347 (Naseem et al., 2026). Detection is close to saturated. Dimension classification is not, and that is where we aim.

4 Related Work

POLAR systems. Large generative models dominated the task. Qwen was used by 31% of teams, LLaMA by 20%, Gemma or Gemini by 19%, and BERT-style encoders by only about 7%. Both English subtask winners are 12B to 27B models (Naseem et al., 2026; Tangkulung and Tsuruoka,

2026; Lin et al., 2026). Among the encoder-based systems, Daniyal et al. (2026) compare BERT variants and Tran (2026) add an LLM refinement stage on top of DeBERTa. Orsanigo et al. (2026) report multi-task gains over single-task training, and Nestor et al. (2026) use label-wise attention. Our evaluation protocol follows Bokaei et al. (2026), who use five-fold cross-validation with three seeds and out-of-fold calibration.

Hierarchy and consistency. Almost every team tunes per-label thresholds (Naseem et al., 2026), so we do not count that as a contribution. On the hierarchy, Guo and Chang (2026) observe violations but leave measurement and correction for later work, and Lin et al. (2026) patch them heuristically at inference. Mohammad (2026) route inputs through a binary gatekeeper, giving non-polarized posts all-zero child labels and forcing an argmax when a polarized post fires nothing. Maharjan et al. (2026) formalise the Logical Violation Rate, audit it across all 22 languages, and enforce a deterministic taxonomic constraint at inference.

The gap. In every one of these systems the constraint acts on outputs. Detection stays an independent prediction that gets overruled later, and Maharjan et al. (2026) go further still by inserting a stop-gradient so the subtasks cannot influence each other’s representations at all. They all report or display the same trade-off, buying consistency with recall.

We read all 67 Task 9 system papers and found none in which the detection decision is a function of the dimension decisions. That reformulation would make violations unrepresentable while discarding nothing, and whether it removes the trade-off is our question. The LVR of Maharjan et al. (2026) serves as the measure, extended to the POLARDETECT/POLARTYPE pair and to both directions of violation.

Foundations. Multi-task learning has a long history in NLP (Caruana, 1997), including learned loss balancing through homoscedastic uncertainty (Kendall et al., 2018). Label-wise attention goes back to Mullenbach et al. (2018); we use it as an established component and claim no novelty for it. Constraining label hierarchies inside neural classifiers is also well established (Giunchiglia and Lukasiewicz, 2020), though nobody has applied it to POLAR.

5 Proposed Methodology

The encoder is DeBERTa-v3-base (He et al., 2023), 184M parameters, which produces contextual token representations H .

5.1 Label-aware dimension detectors

For each of the four named dimensions we learn a query embedding e_k , initialised from the encoder’s representation of the dimension name. Cross-attention between e_k and H gives a label-specific vector h_k , and a linear layer turns this into a logit z_k with probability $p_k = \sigma(z_k)$. Each dimension can therefore attend to different evidence in the input.

Other is handled differently. It is a residual category with no semantics to seed a query embedding from, and other teams have found it difficult (Lin et al., 2026). Its logit z_o therefore comes from the pooled sequence representation instead of a label-name query, so that it captures “polarized, but by none of the named dimensions” instead of pretending to be a fifth topic.

5.2 Structured detection via noisy-OR

Rather than learning a separate detection head, we define the polarization probability as a noisy-OR over the five dimension probabilities:

$$P(pol) = 1 - (1 - p_o) \prod_{k=1}^4 (1 - p_k). \quad (1)$$

A smooth logsumexp over the dimension logits works equally well. Under this parameterisation the model predicts polarized exactly when some dimension fires, and neither of the two impossible outcomes, polarized with no dimension and non-polarized with a dimension, can occur. Consistency comes from the architecture, not from a decoding rule.

5.3 Training objective

POLARTYPE labels are undefined for non-polarized text, so the type loss is masked to polarized instances P :

$$L = L_{detect} + \lambda \cdot \frac{1}{|P|} \sum_{i \in P} L_{type}^{(i)}. \quad (2)$$

Masking matters here. Mohammad (2026) found that training the type head on everything hurt performance, because the all-zero targets dominate the gradient. Eq. 1 is what makes masking safe for us:

on non-polarized instances the detection loss still back-propagates through the noisy-OR and pushes every dimension logit down. The structure replaces the supervision masking takes away, which soft multi-task learning cannot do. For λ we compare a swept constant against learned uncertainty weighting (Kendall et al., 2018), with weighted BCE for the rarer dimensions.

5.4 Experimental models

Table 1 sets out the ladder. M1 reproduces the independent setup of Guo and Chang (2026). M2 is conventional multi-task learning. M3 reimplements the deterministic constraint of Maharjan et al. (2026), using their released code so that the baseline is faithful. M4 is our structured model.

The comparison that matters is M3 against M4, which separates repairing inconsistency from preventing it. Both reach zero violations. The real question is what each one costs to get there. Our expectation is that M3 reproduces the recall loss Maharjan et al. (2026) report while M4 avoids it, since M4 discards no predictions. Should M4 match M3 on consistency and M1 and M2 on recall, the trade-off comes from placement and not from the constraint. Two prior results suggest weight sharing alone will not do it: Guo and Chang (2026) report negative transfer on sparse labels, and Le and Phung (2026) found that initialising the type head from detection weights barely changed anything (0.8316 against 0.8312).

Model	Role
TF-IDF + LogReg	Classical baseline
BERT-base	Transformer baseline
RoBERTa-base	Encoder comparison
M1: Independent heads	Separate models
M2: Shared encoder (soft MTL)	Standard MTL
M3: M2 + deterministic gating	Published fix
M4: Structured (noisy-OR)	Proposed

Table 1: Planned models. M1–M4 form the main ladder, each rung adding one mechanism on the same DeBERTa-v3-base encoder. M1 follows Guo and Chang (2026), and M3 reimplements Eq. 6 of Maharjan et al. (2026). Two further variants of M4, one without label-aware attention and one with Other treated symmetrically, isolate those components.

6 Dataset

The official English subset of POLAR gives us 3,222 training, 160 development, and 1,452 test

instances (Naseem et al., 2026).

Test-label availability. Evaluation ran through Codabench and the gold test labels were never released. If a post-evaluation phase is still open we will submit our final configuration once for a leaderboard-comparable number. If not, every result comes from cross-validation on the training split, and we will say so plainly rather than imply a comparison that does not hold.

Why cross-validation regardless. The development set has 160 instances. Spread across five dimensions, the rarer classes end up with single-digit positive counts, so macro-F1 differences on that split are mostly noise and cannot support ablation claims. Following Bokaei et al. (2026) we run five-fold cross-validation over the 3,222 training instances with five seeds per configuration, and keep the development set as a sanity check.

7 Evaluation Plan

7.1 Metrics

Macro-F1 is the primary metric, with per-dimension F1, precision, and recall alongside it, all given as mean $\pm$ standard deviation over folds and seeds. Adjacent rungs of the ladder are compared using bootstrap resampling (Koehn, 2004).

We additionally report the Logical Violation Rate of Maharjan et al. (2026), adapted to our sub-task pair and made symmetric:

$$LVR = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[\hat{y}_{det}^{(i)} \neq \mathbf{1} \left[\sum_k \hat{y}_k^{(i)} > 0 \right] \right], \quad (3)$$

This is the fraction of predictions where detection contradicts the dimension decisions. Maharjan et al. (2026) count only manifestations predicted for non-polarized text; our version counts both directions, including the case Mohammad (2026) patch with a forced argmax. M3 and M4 will both reach $LVR = 0$, so what we are really measuring is the accuracy each gives up to get there.

Lastly we report macro-F1 per parameter, set against the published English scores of 0.825 for POLARDETECT and 0.532 for POLARTYPE, both obtained with 12B-parameter models.

7.2 Research questions

- **RQ1:** Does deriving detection from the dimension logits improve macro-F1 over independent models (M1) and soft multi-task learning (M2)?

- **RQ2:** Maharjan et al. (2026) report that deterministic gating buys consistency at the cost of multi-label recall. Does that cost come from enforcing the hierarchy, or from enforcing it at inference? We compare M3 and M4 at equal consistency, with both at $LVR = 0$.

- **RQ3:** Do label-aware representations help, and do they help most on instances carrying several dimensions? We evaluate multi-label and single-label instances separately, since an aggregate score hides the difference.

- **RQ4:** How does a 184M-parameter structured encoder compare, per parameter, with the 12B–27B systems leading the English leaderboard?

7.3 Outcome analysis

Each question is worth answering whichever way it comes out. If M4 gets close to the published LLM scores, structure is doing work that scale usually does. If not, we have at least measured the gap. And if M4 cannot recover the recall M3 gives up, the trade-off Maharjan et al. (2026) report belongs to the constraint itself rather than to where it is applied.

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